

12 Iron is a suitable catalyst for the industrial synthesis of ammonia.

Consider the two statements.

- 1 Iron is a good catalyst for this reaction because it donates its 3d electrons to form stronger bonds with the reacting molecules compared to those bonds within the reacting molecules.
- 2 Iron is a good catalyst for this reaction because it forms more than one stable oxidation state.

Which statements are correct?

- A** Neither statement is correct.
- B** Statement 1 only is correct.
- C** Statement 2 only is correct.
- D** Both statements are correct.